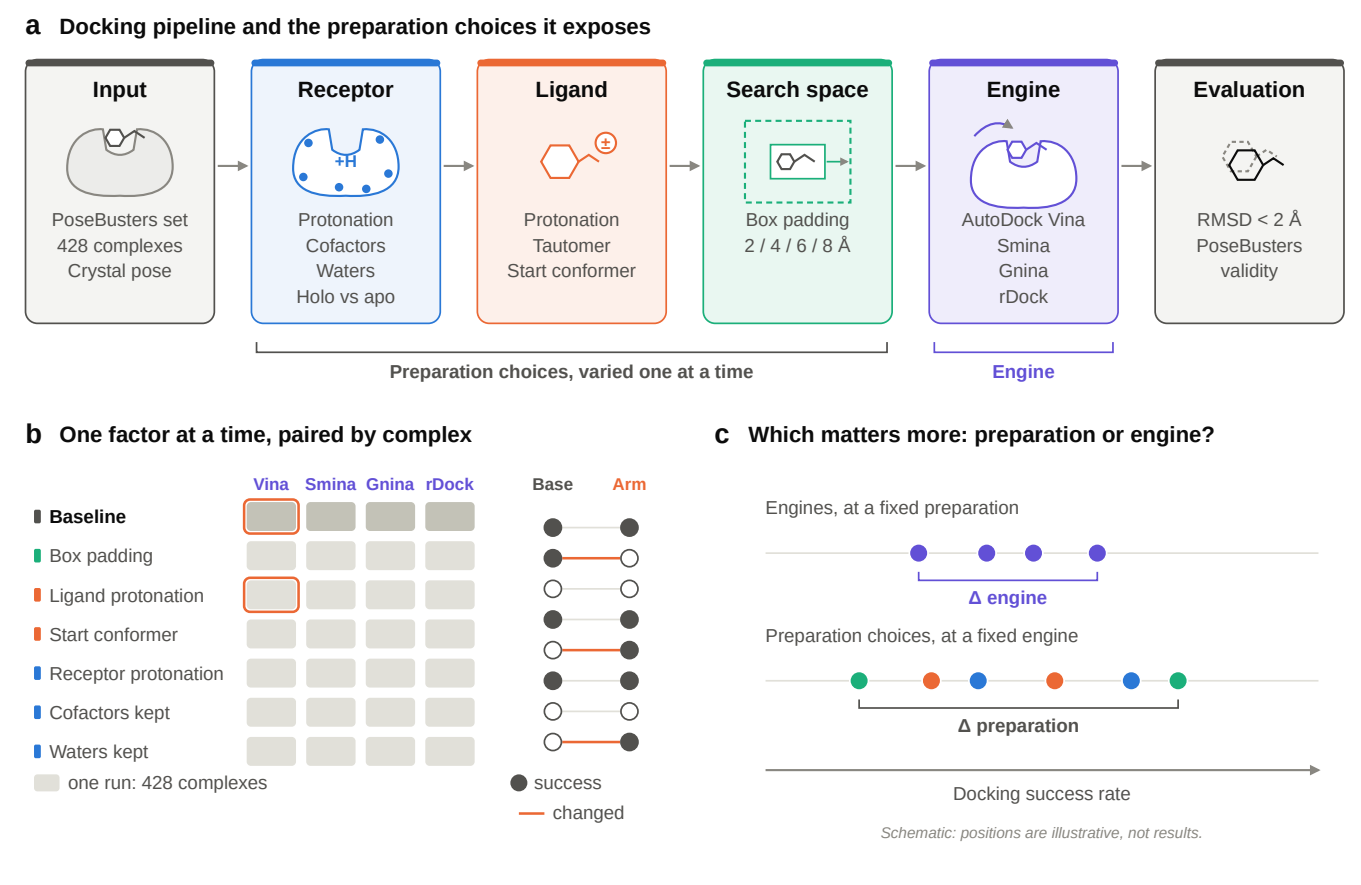
a Docking pipeline and the preparation choices it exposes
Input
PoseBusters set
428 complexes
Crystal pose
Receptor
+H
Protonation
Cofactors
Waters
Holo vs apo
Ligand
±
Protonation
Tautomer
Start conformer
Search space
Box padding
2 / 4 / 6 / 8 Å
Engine
AutoDock Vina
Smina
Gnina
rDock
Evaluation
RMSD < 2 Å
PoseBusters
validity
Preparation choices, varied one at a time
Engine
b One factor at a time, paired by complex
Vina
Smina
Gnina
rDock
Baseline
Box padding
Ligand protonation
Start conformer
Receptor protonation
Cofactors kept
Waters kept
one run: 428 complexes
Base
Arm
success
changed
c Which matters more: preparation or engine?
Engines, at a fixed preparation
Δ engine
Preparation choices, at a fixed engine
Δ preparation
Docking success rate
Schematic: positions are illustrative, not results.